

# Pendulum Lab Certified Chaotic Dynamics Workbench

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Flagship: Melnikov threshold vs period-doubling onset gap map.

Reviewer path: npm run validate:gpu-scale; npm run validate:webgpu-hardware; npm run flagship:certify; npm run flagship:external; npm run reviewer:kit.

Trust model: every headline number carries source, params, uncertainty, reproduce command, artifact hash, and caveat.

GPU rule: 4D and N-chain accelerated results must match CPU f64 or fail closed; the vendor matrix never simulates missing hardware.

Release surfaces: Pages reviewer console, npm OIDC provenance, Zenodo deposition API, SLSA/SBOM attestations, and paper PDF.

Reviewer manifest hash: 0f727f3f087da8

Scorecard hash: 06acb5a49de45e

Flagship certification hash: 1f342d2cb314d6